Three-colour solutions from four-colour cubes

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Abstract

For every $n \geq 2$, any $16n - 3$ cubes, each using all four colours of a fixed palette, contain an $n \times n \times n$ frame whose exterior faces are monochromatic in exactly three colours; this bound is sharp. For a fixed palette, exactly two collections of $16n - 4$ cubes fail, and reflection exchanges them. One cube below this size, exactly 32 collections fail; twenty-four are inclusion-maximal. The threshold for a complete cube is $\max\{n^3, 16n-3\}$; in particular, any n^3 source cubes suffice when $n \geq 4$. The case $n = 2$ answers the threshold question in Problem 2 of Berkove, Eickemeyer and Schulman (2018). The proof combines their corner-containment reduction with orientation constraints and double counting.

1 Introduction

Berkove, Eickemeyer and Schulman (BES) [1, Problem 2] ask for the least number of four-colour cubes that guarantees a choice of eight cubes forming a three-colour corner solution: a $2 \times 2 \times 2$ cube whose exterior faces are monochromatic and use exactly three colours. As in BES [1, Section 1], every source cube uses all four colours of one fixed palette; every target uses exactly three of them. A *variety* is a face colouring up to proper rotation. Collections are multisets of varieties, so repeated copies are allowed.

We prove a sharp threshold for every side length. An n-frame consists of the corners and edges of an $n \times n \times n$ cube, using $12n - 16$ pieces. The least number of source cubes guaranteeing a three-colour n-frame is $16n - 3$ (Theorem 4). Exactly two collections of $16n - 4$ cubes fail: $4n - 1$ copies of each of four varieties, one with a monochromatic corner of each colour, all in the same orientation class. Reflection or any odd colour permutation exchanges these collections. Figure 1 shows the first case, $n = 2$. We also classify all 32 failures of size $16n - 5$, including twenty-four that cannot be enlarged without becoming solvable (Theorem 5).

Filling the face interiors gives the complete-cube threshold $\max\{n^3, 16n - 3\}$ (Corollary 3). In particular, any 64 cubes, each using all four colours of a fixed palette, can all be assembled into a $4 \times 4 \times 4$ cube with monochromatic faces using exactly three colours. Four is the smallest side length for which this guarantee holds without spare cubes; it holds for every larger side length as well.

For comparison, BES [1, Theorem 5] show that any $12n - 16$ four-colour cubes build a four-colour n-frame when $n \geq 3$. Requiring exactly three exterior colours raises the sharp threshold to $16n - 3$.[1] Their completion argument [1, Section 1] gives a four-colour complete cube from any n^3 sources when $n \geq 3$; our result guarantees exactly three exterior colours when $n \geq 4$.

The classification explains the form $16n - 4 = 4(4n - 1)$. Mixing orientation classes allows new corner matches, while repeated-colour demands limit the capacity of each colour bucket. Together these restrictions force an extremal collection into one orientation class and then force the four multiplicities to be equal. We first prove the corner case, including its restricted-target corollary, and then extend the argument to frames.

[1] BES's Table 7 also records the frame number $16n - 17$, for $n \geq 4$, for three-colour source cubes building three-colour frames: a different problem with the same leading coefficient.

2 Model and geometric facts

Reflections are not allowed as moves. Only exterior faces must match the target; internal faces are unrestricted.

A corner type is a cyclically ordered triple of face colours, read counterclockwise when viewed from outside along the corner's body diagonal. Equivalently, order the three outward face normals to form a positively oriented basis. A trichromatic corner has three distinct colours and two orientations, abc and acb. BES [1, Section 2] use clockwise order; our convention reverses every triple, preserving containment and matching. A source cube fits a target corner by a proper rotation if and only if it has the same cyclic corner type: the cyclic alignments of two corner frames are exactly those realised by proper rotations of the cube. For the corner problem ($n = 2$), a target is represented by its multiset of eight corner types. A collection builds a corner solution modelled on that target exactly when these eight occurrences can be matched to distinct cubes having the required types.

For distinct colours a, b, c, write $Q(a; b, c)$ for a target whose b and c faces are opposite and whose other four faces have colour a. Its corner multiset is $4aab + 4aac$. Write $R(a; b, c)$ for the analogous target with b and c adjacent; its corners are

$$2aaa + 2aab + 2aac + abc + acb.$$

Finally, write $D(a, b, c)$ for the target with opposite pairs aa, bb, cc; it has four corners of each orientation on $\{a, b, c\}$. These three families suffice for the corner argument. Section 5 introduces one further target for larger frames.

If every corner type of a variety v occurs on a variety w, replacing a copy of w by v cannot create a solution to any target. In particular, it preserves unsolvability. Table 1 records the reduction. Face tuples use the order $+x, -x, +y, -y, +z, -z$. The rows represent the seven orbits under colour permutations, with rotations already identified. They cover all 68 varieties; four orbits are retained.

Representative	Orbit size	Retained type	Replacement
001123	6	checkerboard C	—
010203	8	monochromatic-corner T	—
010213	12	pair type P	—
010223	12	pair type P	—
000123	12	—	001123
001213	12	—	001123
010123	6	—	001123

Table 1: The 68 varieties reduce to 38.

A reduced variety belongs to bucket a if it has at least two of the three corner types aab with $b \neq a$. The reduction, 38 minimal varieties, orientation classes, checkerboards and buckets below are from BES [1, Section 2, Table 8 and Definition 3]; we count each bucket's seven mirror pairs as fourteen individual varieties.

Lemma 1 (Finite geometry)**.** *Every non-retained variety in Table 1 contains the corner types of a checkerboard, so every variety reduces to one of the 38 retained varieties. Their bucket incidences are:*

- *There are eight T varieties. Each has a repeated colour a, corners aaa and aab for every $b \neq a$, and belongs only to bucket a.*
- *The 24 P varieties belong to two buckets, one for each repeated colour.*

- *The six checkerboards belong to no bucket. Their repeated colours occupy opposite face pairs; with repeated colours a, b, a checkerboard has both orientations on $\{a, b, c\}$ and $\{a, b, d\}$.*

The 32 ordinary varieties (T and P) split into two classes of 16, corresponding to BES's K_o and K_m (up to interchanging the names). Every member of a class has exactly the same four trichromatic corner types, one for each three-element colour set. The two classes have opposite orientations. In our convention one class has the trichromatic corner set

$$\Omega = \{012, 031, 023, 132\},$$

one set of BES's universal corners*. Each class contains exactly one T variety for each repeated colour. Even colour permutations preserve the classes; odd colour permutations and reflection exchange them.*

Proof. The face-colour multiplicities are $3+1+1+1$ or $2+2+1+1$. In the first case, the three equal faces either meet at a vertex or contain an opposite pair, giving two orbits. In the second case, both repeated pairs can be opposite, exactly one can be opposite, or both can be adjacent. In the last case, the two monochromatic edges either have perpendicular directions, giving two mirror configurations represented by 010213 and 010223, or are parallel, represented by 010123. These cases cover all varieties. The orbit sizes in Table 1 sum to 68, the total number of varieties [1], so the seven orbits are distinct. The three non-retained representatives contain the corner types of 001123; colour permutations transport these containments to their whole orbits. Table 2 gives the corner sets of the retained representatives. All three ordinary representatives in Table 2 belong to the class of Ω. Even colour permutations preserve Ω and odd ones reverse it, with triples read cyclically. Applying colour permutations to Table 2 gives the stated incidences and orientation classes. The eight T varieties split evenly between the classes, and even permutations act transitively on the four colours, so each class contains exactly one T variety for each repeated colour. Reflection reverses every trichromatic corner. □

Representative	Corner-type set
001123	$\{012, 013, 021, 031\}$
010203	$\{000, 001, 002, 003, 012, 023, 031, 132\}$
010213	$\{001, 003, 011, 012, 023, 031, 112, 132\}$
010223	$\{002, 003, 012, 022, 023, 031, 122, 132\}$

Table 2: Corner-type sets of the retained representatives, with cyclic triples written in lexicographically least form.

Lemma 2 (Bucket bound)**.** *Eight reduced cubes in bucket a can build a target $Q(a; b, c)$ for some b, c. Consequently, a reduced collection with no Q target has at most seven cubes in each bucket.*

Proof. Each of the eight cubes misses at most one of the three types aab. Thus at most one type is missed by five or more cubes. Choose two types, aab and aac, each missed by at most four cubes. Assign cubes missing aac to aab positions, and cubes missing aab to aac positions. These groups are disjoint and each has size at most four. Every remaining cube has both types, so they fill the remaining positions of $Q(a; b, c)$. □

3 The extremal collections

Theorem 1. *A collection of 28 four-colour cubes that builds no target of type Q, R or D consists of seven copies of each of the four T varieties in one ordinary orientation class.*

Proof. Replace each non-retained cube by a checkerboard using Lemma 1, leaving retained cubes unchanged. This preserves avoidance of all three target families. Let $A \geq B$ count the cubes belonging to the two ordinary orientation classes, and let C count checkerboards, so $A + B + C = 28$. Call the class with A cubes the majority class (either one if equal). Call the remaining $M = B + C$ cubes *minority cubes*. A minority cube *supports* a three-element colour set E if it has the orientation on E opposite to the majority class. Smaller-class cubes support every E; a checkerboard supports two.

(i) At most six minority cubes: $M \leq 6$. For $E = \{a, b, c\}$, let z_E count checkerboards supporting E. If $z_E \geq 8$, those eight checkerboards build the target $D(a, b, c)$. Thus $z_E \leq 7$ for each E, and, since each checkerboard supports two triples, $2C \leq 28$. It follows that $A + B \geq 14$, so $A \geq 7$. There are $m_E = B + z_E$ minority cubes supporting E. If $m_E \geq 4$, four such cubes and four majority cubes build $D(a, b, c)$. Hence $m_E \leq 3$ for every E. Summing gives

$$4B + 2C \leq 12, \qquad M \leq 6, \qquad A = 28 - M \geq 22. \tag{1}$$

(ii) No minority cubes: $M = 0$. Let t_a and p_a count only majority-class T and P cubes in bucket a, and let n_T, n_P be their respective totals, so $A = n_T + n_P$. By Lemma 2, $t_a + p_a \leq 7$. Each T cube belongs to one bucket and each P cube to two. This is BES's double-count [1, equation (1)]; summing the bucket bounds and using $n_T + n_P = 28 - M$ gives

$$n_T + 2n_P \leq 28, \qquad n_P \leq M. \tag{2}$$

We use the following construction. If six majority cubes supply $2aaa + 2aab + 2aac$ and a minority cube supports $\{a, b, c\}$, then the minority cube supplies the opposite trichromatic orientation and an unused majority cube supplies the other. Such a cube is available since $A \geq 22$. This builds $R(a; b, c)$.

Suppose $M > 0$. We claim that $t_a \geq 4$ forces $t_a + p_a \leq 5$. Otherwise choose six majority cubes in bucket a, including at least four T cubes. Any minority cube supports some triple containing a, say $\{a, b, c\}$, since a checkerboard's two supported triples together contain every colour. Each of the at most two P cubes has two of the three types aax, so it has aab or aac; since $R(a; b, c)$ needs two of each, each P cube can take a position of a type it has. The T cubes have aaa and every type aax, so they fill the remaining positions of $2aaa + 2aab + 2aac$, and the construction above gives a solution, a contradiction. In particular, $t_a \leq 5$ for every a.

If $t_a \leq 3$, the bucket bound gives $2t_a + p_a \leq t_a + 7 \leq 10$. If $4 \leq t_a \leq 5$, the claim gives $2t_a + p_a \leq t_a + 5 \leq 10$. Thus $2t_a + p_a \leq 10$ in all cases. Each majority T cube contributes two through its single bucket, and each majority P cube contributes one through each of its two buckets. Summing gives

$$2A = 2n_T + 2n_P = \sum_a (2t_a + p_a) \leq 40,$$

contradicting $A \geq 22$ from (1). Therefore $M = 0$.

(iii) Equal multiplicities: $t_a = 7$. With $M = 0$, (2) forces $n_P = 0$. All 28 cubes are majority-class T cubes, and the four bucket bounds force each $t_a = 7$.

Since the reduced collection contains no checkerboards, no replacement occurred: the original collection was already reduced. □

Theorem 2. *Every collection of 29 four-colour cubes contains a three-colour corner solution.*

Proof. Otherwise every 28-cube subcollection would have the form in Theorem 1. Take one such subcollection S and let v be the remaining cube. Remove from S a cube of a variety different from v, and insert v. The multiplicity of v's variety is now 1 or 8, whereas every variety in a collection described by Theorem 1 has multiplicity 0 or 7, a contradiction. □

4 Sharpness

Lemma 3. *In a cube using exactly three colours, the two orientations of a trichromatic corner occur equally often. If no trichromatic corner occurs, the cube is $Q(a;b,c)$ for some distinct a,b,c.*

Proof. For any cube coloured from a palette $\{a,b,c,d\}$, let $\nu(xyz)$ count corners of cyclic type xyz. Read the three directed transitions around each corner in the chosen cyclic order. Each edge between an a face and a b face contributes ab at one endpoint and ba at the other, so the total counts of these transitions agree. Corners with repeated colours contribute equally to both counts. The remaining contributions give

$$\nu(abc) - \nu(acb) + \nu(abd) - \nu(adb) = 0. \tag{3}$$

If only a,b,c occur, the last two terms vanish, proving the balance.

For the second assertion, if a colour c occurs on only one face, its four neighbours must all have the same colour a: a change from a to b around this face would produce a trichromatic corner. The opposite face is then the sole b face. Otherwise each colour occurs twice. If the two c faces are opposite, the four neighbours of either contain both a and b, forcing a change. If they are adjacent, the other three neighbours form a path containing both a and b, since each occurs only twice. Again a change is forced. Both cases produce a trichromatic corner. □

Theorem 3. *The threshold 29 in Theorem 2 is best possible. For a fixed palette, exactly two 28-cube collections fail, and they are mirror images. Up to colour permutation, there is a single obstruction.*

Proof. Take seven copies of each of the following four face tuples (Figure 1):

$$010203, \qquad 011213, \qquad 021223, \qquad 031323.$$

These are T cubes with repeated colours $0,1,2,3$, respectively, all in the same ordinary orientation class. A target with any trichromatic corner requires both orientations by Lemma 3, whereas the collection supplies only one. A target without one is $Q(a;b,c)$; every one of its eight corners has colour a twice, so only the seven T cubes of repeated colour a are usable. No three-colour target is possible. Reflection gives the analogous example in the other orientation class. Theorem 1 excludes all other collections. Every odd colour permutation exchanges the two orientation classes and hence these two collections. □

Remark (Allowing fewer colours). If targets may use at most three colours, 28 cubes suffice. In either exceptional collection, four T cubes of repeated colour a and four of repeated colour b supply $aaa + 3aab + bbb + 3bba$. These are the corners of a two-colour cube whose three a faces meet at one vertex and whose three b faces meet at the opposite vertex. Every other 28-cube collection builds a three-colour target by Theorem 3. This gives an upper bound, not an exact threshold, for that variant.

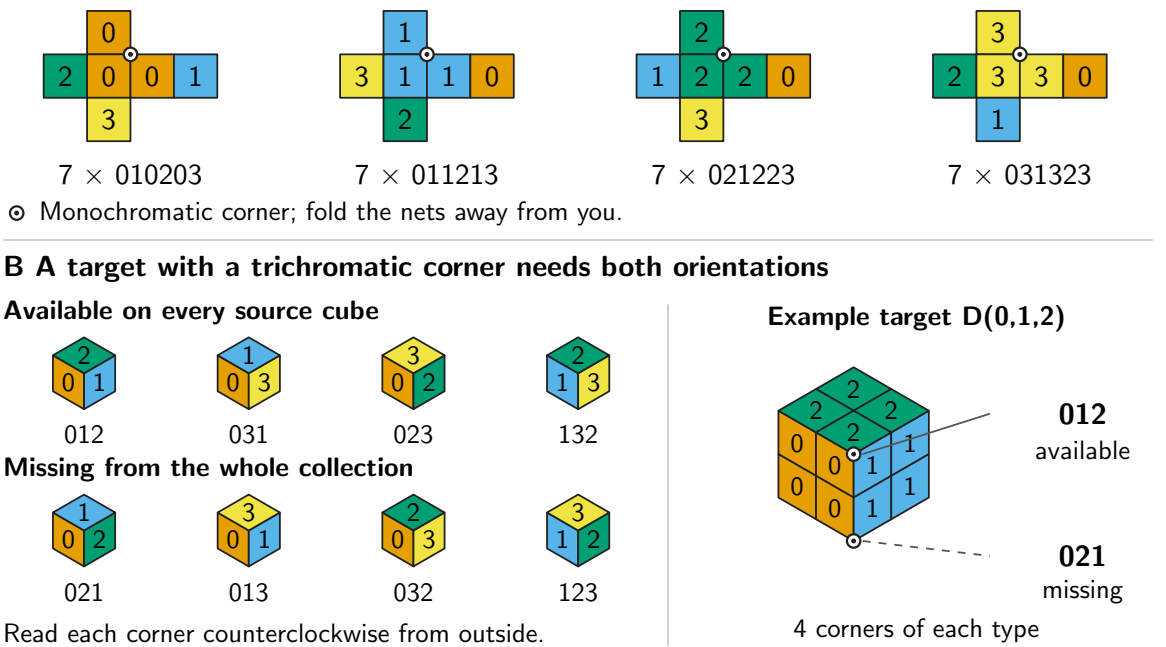


Figure 1: Why one exceptional 28-cube collection fails for three-colour targets. (A) Seven copies of each variety, with monochromatic corners marked. (B) Targets with a trichromatic corner need both orientations; the example's bottom face is hidden. (C) Targets without trichromatic corners require eight cubes with the same repeated colour, but only seven are available; a, b, c are distinct. Reflection gives the other obstruction.

Corollary 1 (Restricted targets)**.** *Every collection of 29 four-colour cubes, and every solvable collection of 28, builds a target of type Q, R or D. These families contain just 28 of the 120 target varieties.*[2]

Proof. By Theorems 1 and 3, a 28-cube collection avoiding these families is unsolvable. The swap in the proof of Theorem 2 rules out a 29-cube collection avoiding them. There are $4\binom{3}{2} = 12$ varieties each of Q and R: choose a and the unordered pair $\{b, c\}$. There are $\binom{4}{3} = 4$ varieties of D, one for each three-colour set. □

[2]For any fixed three colours, a rotation with k face cycles fixes $3^k - 3 \cdot 2^k + 3$ colourings using all three. The 24 rotations have $6, 4, 3, 2$ face cycles with multiplicities $1, 3, 12, 8$, so Burnside's lemma gives $(540+3\cdot36+12\cdot6)/24 = 30$ varieties. Four choices of omitted colour give 120.

Remark. For the palette $\{0,1,2,3\}$, write $\delta(xyz) = \nu(xyz) - \nu(xzy)$. Applying (3) to the different colour pairs gives

$$\delta(012) = \delta(031) = \delta(023) = \delta(132).$$

This common integer is the cube's *degree*, relative to Ω. Reflection negates the degree, as does each transposition of colour labels; thus even colour permutations preserve it and odd ones negate it. Each ordinary representative in Table 2 has eight distinct corner types, so each occurs once; the two ordinary classes therefore have degrees $+1$ and -1. The checkerboard representative has neither 023 nor 032, so its degree, and that of every checkerboard, is zero.

5 Frames of arbitrary side length

An n-frame consists of the eight corner cubes and the $n-2$ further cubes on each of the twelve edges of an $n \times n \times n$ cube. Thus it contains $L = 12n - 16$ cubes. Its six exterior faces are monochromatic and use exactly three colours; face interiors are not part of the frame. This is the frame model of BES [1, Section 1], with three target colours instead of four.

Theorem 4 (Frame threshold and classification)**.** *For every $n \geq 2$, every collection of $16n-3$ four-colour cubes contains a three-colour n-frame. For a fixed palette, exactly two collections of $16n-4$ cubes fail: $4n-1$ copies of each of the four T varieties in one orientation class, or their reflections. Consequently the sharp frame threshold is $F(n) = 16n-3$.*

The corner-containment reduction also preserves edge availability: an edge occurs in a corner, so containment of corner types implies containment of edge types [1, Section 2, after Definition 2]. An edge type is an unordered pair of colours, allowing repetition. Every ordinary variety has all six bichromatic edge types. A checkerboard has five, missing precisely the pair of its singleton colours; different checkerboards miss different pairs. These facts follow from the representatives in Table 2, and also appear in BES's frame arguments [1, proofs of Lemmas 7–8 and Table 5]. A reduced cube has an aa edge if and only if it belongs to bucket a.

Lemma 4. *Let $n \geq 2$. If a reduced collection of at least $16n-5$ cubes has a D corner solution, it has an n-frame.*

Proof. If a checkerboard variety occurs at least L times, those copies build a D-frame on either supported triple. Otherwise every checkerboard variety occurs at most $L-1$ times. Remove the eight cubes in the given $D(a,b,c)$ corner solution. The remaining positions demand $4(n-2)$ copies of each of ab, ac, bc. Each type is available on at least

$$(16n-5) - 8 - (L-1) = 4n+4$$

remaining cubes. The union of the available source sets for any two types is the entire remaining collection, since no reduced cube misses two bichromatic types. There are at least $12(n-2)$ remaining cubes. These are Hall's conditions for the three groups of edge positions, so all positions can be filled with distinct copies. □

One more target supplies the growing repeated-colour demands. For distinct a, b, c, let $U(a,b;c)$ have face tuple (a,b,a,b,a,c). Its corner and edge multisets are respectively

$$aaa + 2aab + aac + abb + bbc + abc + acb, \qquad 3aa + bb + 4ab + 2ac + 2bc.$$

Write T_a for the T variety with repeated colour a in a given ordinary class. Thus an ordinary class supplies a U-frame using $3n-2$ of its T_a cubes, n of its T_b cubes, and $1 + 8(n-2)$ other cubes, together with one minority cube supporting $\{a,b,c\}$. The T_a cubes fill four corners and the $3(n-2)$ monochromatic aa edge positions; the T_b cubes fill two corners and the $n-2$

monochromatic bb positions. The minority cube and one further cube from the same ordinary class fill the two trichromatic corners. Every remaining edge is bichromatic and can use any ordinary cube.

Proof of Theorem 4. We first classify a failing collection of $N = 16n - 4$ cubes. The case $n = 2$ is Theorem 3, so assume $n \geq 3$. Apply the same reduction as in Theorem 1, replacing every non-retained variety by a checkerboard. In the reduced collection, let $A \geq B$ count the cubes in the two ordinary classes, let C count checkerboards, and put $M = B + C$. Define t_a, p_a, n_T, n_P for the majority class as before.

By Lemma 4, no D corner solution exists. The first step of Theorem 1 uses only these targets, so it still applies, since $N \geq 28$ preserves its preliminary bound $A \geq 7$. It gives

$$4B + 2C \leq 12, \qquad M = B + C \leq 6, \qquad A = N - M \geq 16n - 10.$$

A majority bucket cannot contain $4n$ cubes: eight build a Q corner solution by Lemma 2, another $4(n-2)$ fill its aa edge positions, and $8(n-2)$ further majority cubes fill its bichromatic edges. The supply is sufficient since $A - 4n \geq 12n - 10 \geq 8(n-2)$. Consequently

$$t_a + p_a \leq 4n - 1, \qquad n_T + 2n_P \leq 16n - 4.$$

Together with $n_T + n_P = N - M$, this implies

$$n_P \leq M, \qquad n_T \geq 16n - 16.$$

Suppose $M > 0$. Some colour a has $t_a \geq 4n - 4 \geq 3n - 2$. At most one colour has fewer than n majority T cubes: two such colours would give

$$n_T \leq 2(n-1) + 2(4n-1) = 10n - 4 < 16n - 16.$$

Choose a minority cube. It supports a triple containing a: smaller-class ordinary cubes support every triple, and a checkerboard's two supported triples together contain every colour. At least one of the other two colours in that triple has at least n majority T cubes; call it b, and call the third c. The $U(a, b; c)$ construction now applies. After selecting its $4n - 1$ majority corner and monochromatic-edge cubes, at least $A - (4n-1) \geq 12n - 9 \geq 8(n-2)$ majority cubes remain. This contradicts failure.

Hence $M = 0$, then $n_P = 0$, and the four bucket capacities force $t_a = 4n - 1$ for every a. No checkerboards remain, so no replacement occurred. This proves the classification of possible failures.

Both stated collections do fail. A target with a trichromatic corner needs both orientations by Lemma 3. Otherwise it is $Q(a; b, c)$. Its eight corners and its $4(n-2)$ monochromatic aa edge positions all require T_a cubes, giving demand $4n$ and supply $4n - 1$.

Finally, a failing collection of $16n-3$ cubes would have every $(16n-4)$-cube subcollection of the classified form. The swap in Theorem 2 changes one variety's multiplicity to 1 or $4n$, whereas the classification allows only 0 and $4n - 1$. This contradiction proves the upper bound. □

Theorem 5 (One cube below the extremal size)**.** *For every $n \geq 2$, exactly 32 fixed-palette collections of $16n - 5$ cubes fail to contain a three-colour n-frame. Put $h = 4n - 1$. They are:*

1. *the eight collections*
$$(h-1)T_a + hT_b + hT_c + hT_d,$$
one for each choice of ordinary orientation class and colour a;

2. *the twenty-four collections*
$$P + (h-1)T_a + (h-1)T_b + hT_c + hT_d,$$
one for each ordinary P variety, where a, b are its bucket colours and the four T varieties belong to its orientation class.

Each collection in the second family is inclusion-maximal among failures: adding any cube makes it solvable.

Proof. Let $N = 16n - 5$ and retain the majority notation from the frame proof. For $n \geq 3$, Lemma 4 and the same orientation count give $M \leq 6$ and $A \geq 16n - 11$. A majority bucket of size $4n$ would still build a Q-frame, so

$$n_T + 2n_P \leq 16n - 4, \qquad n_P \leq M + 1, \qquad n_T \geq 16n - 18.$$

If $M > 0$, some $t_a \geq 4n - 4 \geq 3n - 2$, and at most one colour has $t_b < n$: two such colours would give $n_T \leq 10n - 4 < 16n - 18$. The same U construction applies, with at least $A - (4n - 1) \geq 12n - 10 \geq 8(n - 2)$ ordinary cubes left for its bichromatic edges. Hence $M = 0$.

For $n = 2$, the orientation count at $N = 27$ gives $M \leq 6$ and $A \geq 21$. If $M > 0$, the R argument in Theorem 1 still implies $2t_a + p_a \leq 10$ for every colour. Summing yields $2A \leq 40$, a contradiction. Thus $M = 0$ here too.

Now $n_P \leq 1$. If $n_P = 0$, four bucket capacities of h and total $4h - 1$ leave exactly one T multiplicity at $h - 1$, giving the eight deletion cases. If $n_P = 1$, every bucket must have size h, since $n_T + 2n_P = 4h$. The unique P cube fills two buckets, giving the stated multiplicities and twenty-four cases. No checkerboards remain, so the reduction made no replacements in the original collection.

Conversely, each listed collection lies in one ordinary orientation class. Orientation balance excludes every target with a trichromatic corner. A remaining target $Q(a; b, c)$ requires $4n$ distinct cubes with an aa edge for its corners and monochromatic edges, while bucket a has at most $h = 4n - 1$ cubes. Every listed collection therefore fails. Finally, adding any cube to a pair case produces a collection of size $16n - 4$ still containing a P cube; Theorem 4 excludes it from the failing collections of that size. □

Corollary 2. *For $n \geq 3$, every collection of at least $16n - 3$ source cubes has an n-frame modelled on one of the 40 targets of types Q, D, U.*

Proof. The frame counting argument uses only these three types: avoiding them forces purity and four bucket capacities of $4n-1$, hence at most $16n-4$ cubes. There are twelve Q targets and four D targets. The ordered choice of three distinct colours in $U(a, b; c)$ gives $4 \cdot 3 \cdot 2 = 24$ varieties: their face multiplicities $3, 2, 1$ recover a, b, c, respectively. These families are disjoint. □

Corollary 3 (Complete cubes)**.** *For every integer $n \geq 2$, the sharp threshold for a complete $n \times n \times n$ cube with exactly three exterior colours is*

$$G(n) = \max\{n^3, 16n - 3\}.$$

In particular, $G(2) = 29$, $G(3) = 45$, and $G(n) = n^3$ for $n \geq 4$.

Proof. Both terms are necessary: an assembly uses n^3 pieces, and contains an n-frame. Conversely, construct a frame using Theorem 4. Each face-interior position prescribes just one colour, which every unused source cube has; interior positions prescribe none. Thus arbitrary unused cubes complete the assembly whenever at least n^3 pieces are available. □

6 Formal verification and computation

The frame threshold, both obstruction classifications, complete-cube threshold and restricted-target corollaries are formalised in Lean 4.35.0 [3] with mathlib v4.35.0 [4] for arbitrary indexed families of four-colour cubes. Solutions select distinct copies and proper rotations matching the exposed faces at every required position. The frame statements cover every side length $n \geq 2$.

The threshold and sharpness statements, and the corner restricted-target statements, are formulated without catalogue indices. The forty-target extension uses a canonical target index and proves proper-rotation placements for arbitrary indexed source families. The classifications specify multiplicities of rotation classes. Kernel checks establish the catalogue and finite geometric facts, and a lemma for arbitrary palettes equates cyclic corner matching with physical placement.

Thirty-five results and the required definitions are restated in a standalone verification file (the comparator challenge) importing only mathlib. The near-extremal classification and forty-target extension are additional library theorems, with physical-family statements and axiom audits; they are not yet added to that challenge. Automated checks compare the challenge statements with the proof library and replay the proofs through the Lean, NanoDa and con-ron kernels.

The audit reports three standard axioms: `propext`, `Classical.choice` and `Quot.sound`. No proof uses holes or `native_decide`. The repository[3] provides sources, reproduction commands and details of the statement checks. The physical model and palette-independent geometry come from the shared `colour-cubes-core` library [5]. The challenge restates its definitions locally.

The Lean proof uses counting and finite Hall certificates. Its corner-case proof of the extremal classification treats checkerboards and the smaller ordinary class separately. The near-extremal proof combines them, as does the frame proof. Finite checks establish each target's corner and edge requirements, while the allocation and counting proofs are symbolic in the side length. Neither the written proof nor the formalisation enumerates all source collections.

A computational search found the corner obstruction and agreed with the corner threshold; separate scripts check the geometric tables, incidences, edge constructions and sample frame obstructions. The correctness of these external computations is not assumed in either the written proof or the formalisation.

In the guaranteed range, an assembly can be found by testing just 28 Q, R, D targets when $n = 2$ (Corollary 1), or 40 Q, D, U targets when $n \geq 3$ (Corollary 2), using capacitated bipartite matching. Testing all 120 targets decides feasibility for arbitrary smaller collections; the restricted-target guarantee alone does not do so. Source varieties have capacities equal to their available multiplicities; corner and edge requirements have capacities equal to their demands. A source is adjacent to a requirement when it has the required corner or edge type. Grouping identical requirements keeps the number of vertices and edges bounded independently of the side length; only the capacities vary. An integral flow meeting all demands determines a frame, and compatible proper rotations can then be chosen. Unused cubes complete the assembly whenever at least n^3 pieces are available.

Further questions

An elementary classification of smaller inclusion-maximal failures would extend Theorem 5. Cuboids pose a separate problem: for each $k \geq 1$, take $2k$ copies of each of the six checkerboards. These $12k$ cubes cannot fill a $2 \times 2 \times 3k$ box in exactly three colours. Every target corner must be trichromatic, forcing a D target. Every position then demands the same bichromatic edge type along the long axis, including the corners. One checkerboard lacks that type, so only $10k$ cubes are usable for $12k$ positions. Thus assembly from exactly the volume fails for arbitrarily large cuboids, even though it holds for cubes of side length at least four.

[3] https://github.com/jasondavies/four-to-three-colour-cubes.

Literature and priority

Literature searches on November 1–2, 2026 found no earlier determination of the corner threshold or the frame formula. These searches do not establish priority. Rosenfeld's October 2026 article [2] *MacMahon Cubes and Moshe's Insanity, Puzzle 1* remains unread, and its relevance is undetermined.

Declaration of AI-generated research and review

GPT-6-Astra (OpenAI), used through Codex, developed the original research argument, computational search, Lean formalisation and manuscript under the author's direction. Claude Opus 5.5 (Anthropic), used through Claude Code, contributed the minority-cube argument, balance proof, further mathematical simplifications, illustrations and review. Subsequent Codex revisions incorporated and extended these contributions. Detailed attribution and review scope are recorded in the repository's `formalization.yaml`. The human author initiated and directed the research and accepts responsibility for the paper's content and the accuracy of this disclosure.

References

[1] E. Berkove, A. Eickemeyer and M. Schulman, *The 4-Color Cubes Puzzle*, Integers **18** (2018), A68. Journal PDF.

[2] M. Rosenfeld, *MacMahon Cubes and Moshe's Insanity, Puzzle 1*, Geombinatorics Quarterly **36**(2) (October 2026), starts at p. 78. Journal listing; full text not examined.

[3] L. de Moura and S. Ullrich, *The Lean 4 Theorem Prover and Programming Language*, in Automated Deduction—CADE 28, LNCS **12699**, Springer (2021), 625–635. doi:10.1007/978-3-030-79876-5_37.

[4] The mathlib Community, *The Lean Mathematical Library*, in Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), ACM (2020), 367–381. doi:10.1145/3372885.3373824. arXiv:1910.09336.

[5] J. Davies, `colour-cubes-core`: palette-independent Lean definitions and proofs for face-coloured cubes, Lean library (2026). Release `v1.0.0`, `https://github.com/jasondavies/colour-cubes-core`.